



Cloudflare Radar

2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network.

Source: Cloudflare Radar | radar.cloudflare.com/year-in-review/2024

20
24

17% Global Traffic Growth and Starlink's 3.3x Surge

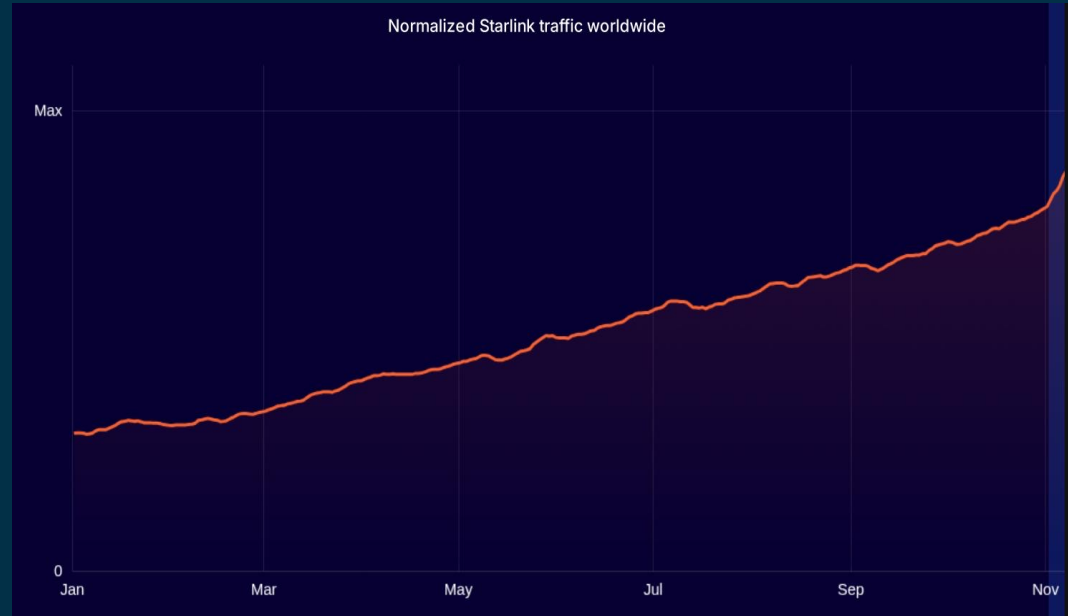
17%

Year-over-year global Internet traffic growth

3.3x

SpaceX Starlink traffic growth in 2024

- Global Internet traffic grew 17% in 2024, continuing the upward trend
- Growth driven by communication, commerce, entertainment, and transportation
- Starlink bringing satellite connectivity to



Normalized Starlink traffic worldwide showing 3.3x growth in 2024

Source: Cloudflare Radar 2024

Google Leads Top 10 Services as Generative AI Rises

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

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Top 10 most popular Internet services worldwide

Key Insight: Google tops the rankings for the second consecutive year. Generative AI continued its rapid rise throughout 2024, while the Metaverse receded. Rankings based on Cloudflare 1.1.1.1 DNS resolver data.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



Post-quantum encrypted traffic worldwide showing rapid adoption after Chrome 124 release

2% → 13%

Post-quantum encryption share of TLS 1.3 traffic after Chrome 124 default enablement (April 17, 2024)

21%

HTTP/3 adoption — providing faster connections, packet-loss mitigation, and encryption by default via QUIC

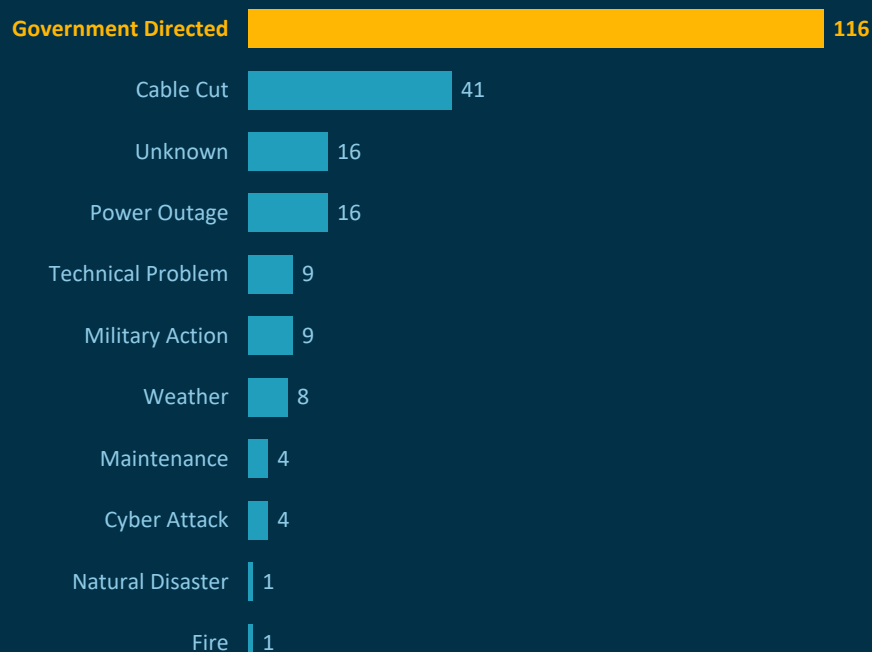
Go ≈ 12%

Top API client language, followed by Node.js, Python (9.6%), Java (7.4%), and .NET (3.6%)

Mozilla Firefox has also started rolling out post-quantum by default; Apple Safari began initial testing. Mobile: 33% iOS traffic share. IPv6: 28% of dual-stack traffic.

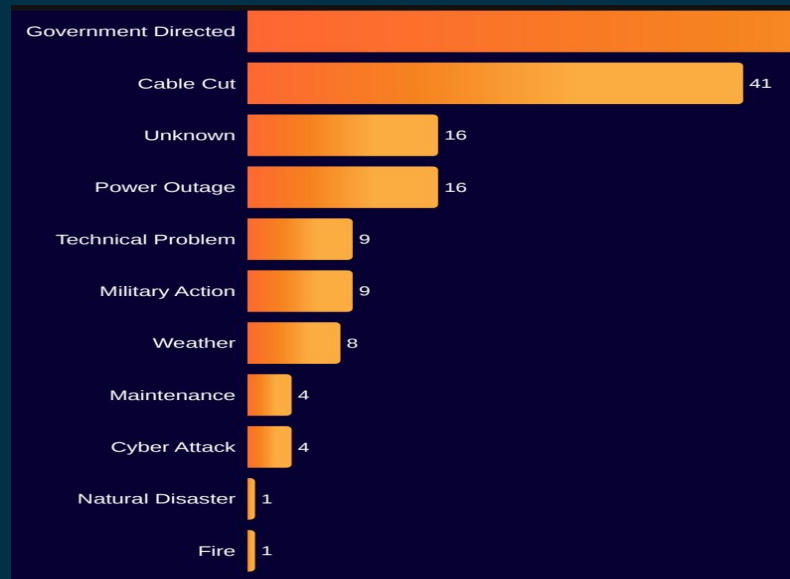
Source: Cloudflare Radar 2024

225 Major Disruptions: Government Shutdowns Lead Outage Causes



225 total major Internet disruptions observed globally in 2024

Source: Cloudflare Radar Outage Center 2024



Causes of major Internet disruptions observed in 2024

41% of global traffic came from mobile devices. Spain recorded the highest download speed among all countries measured.

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

of all traffic mitigated

Scale of ongoing threats processed by
Cloudflare

4.3%

of emails were malicious

43% of malicious emails contained
deceptive links

68%

of bot traffic from top 10

countries
High geographic concentration of
automated traffic

Most Attacked Sector

Gambling & Games

was the most attacked industry category in 2024

Log4j (CVE-2021-44228)

continues to be actively targeted more than two years after disclosure — underscoring the persistent nature of widely known vulnerabilities

Email & Routing Security

43% of malicious emails contained deceptive links

.bar top-level domain originated the largest share of malicious and spam email

RPKI valid routes increased by 6.4%

— an important step toward improved Internet routing security

Five Takeaways Shaping the 2025 Internet Outlook

01

Traffic Growth Trajectory

Global Internet traffic grew 17% year-over-year, with Starlink surging 3.3x as satellite connectivity expands to underserved regions — signaling continued demand for bandwidth and access.

02

Post-Quantum Encryption Momentum

Post-quantum encryption leapt from ~2% to 13% of TLS 1.3 traffic following Chrome 124's default enablement — a paradigm shift in cryptographic standards that organizations must prepare to support.

03

Connectivity Fragility from Government Actions

Government-directed shutdowns were the leading cause of 225 major Internet disruptions (116+ incidents) — highlighting how policy decisions directly impact global connectivity.

04

Persistent Security Threats

6.5% of all traffic was mitigated, Gambling/Games was the most targeted sector, and Log4j exploitation persists years after disclosure — security vigilance remains non-negotiable.

05

Evolving Developer Ecosystem

Go surpassed other languages to become the top API client language at ~12% of automated API traffic, while HTTP/3 reached 21% adoption — the infrastructure stack is actively modernizing.